A Bayesian-Kelly Pareto Frontier for Optimal Transaction Acceptance under Uncertainty *

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Abstract

We characterize the decision boundary for transaction acceptance by synthesizing the Kelly Criterion with Bayesian inference. By applying Kelly criterion we derive a hyperbolic Pareto frontier which converges on a minimalist expected-value requirement ($p \cdot r > c$). By modeling the probability of success through a Beta distribution prior, we show that, for any rational agent, the "Go/No-Go" decision is governed by a simple critical ratio: the net profit must exceed the ratio of observed frictions to successes.

***K*eywords** Bayesian Inference · Kelly Criterion · Pareto Frontier · Decision Theory · Risk Management · Transaction Acceptance

1 The Profit Model

Assume a two-player game where Player A incurs a cost c to engage and stands to gain a reward r. We define net profit b as the ratio of net gain to cost:

$$b = \frac{r - c}{c}$$

2 The Kelly Criterion Constraint

To determine the optimal fraction of capital to risk, we employ the Kelly Criterion (K) [1]. For a transaction to be viable, the expected growth rate must be positive ($K \geq 0$):

$$K = \frac{bp - (1 - p)}{b} \geq 0$$

This inequality yields the immediate threshold for the probability of success:

$$p \geq \frac{1}{1 + b}$$

This is, in fact, just an intuitive requirement: $p \cdot r \geq c$ - simply that the expected reward exceeds the costs of the transaction.

3 Bayesian Probability of Success

Player A treats the transaction as business under uncertainty. Using a Bayesian [2] updating scheme over multiple observations, we model the probability of success p using a Beta distribution $Beta(\alpha, \beta)$, where α represents the counts of successes and β represents failures (frictions). The expected probability is [3]:

$$p = \frac{\alpha}{\alpha + \beta}$$

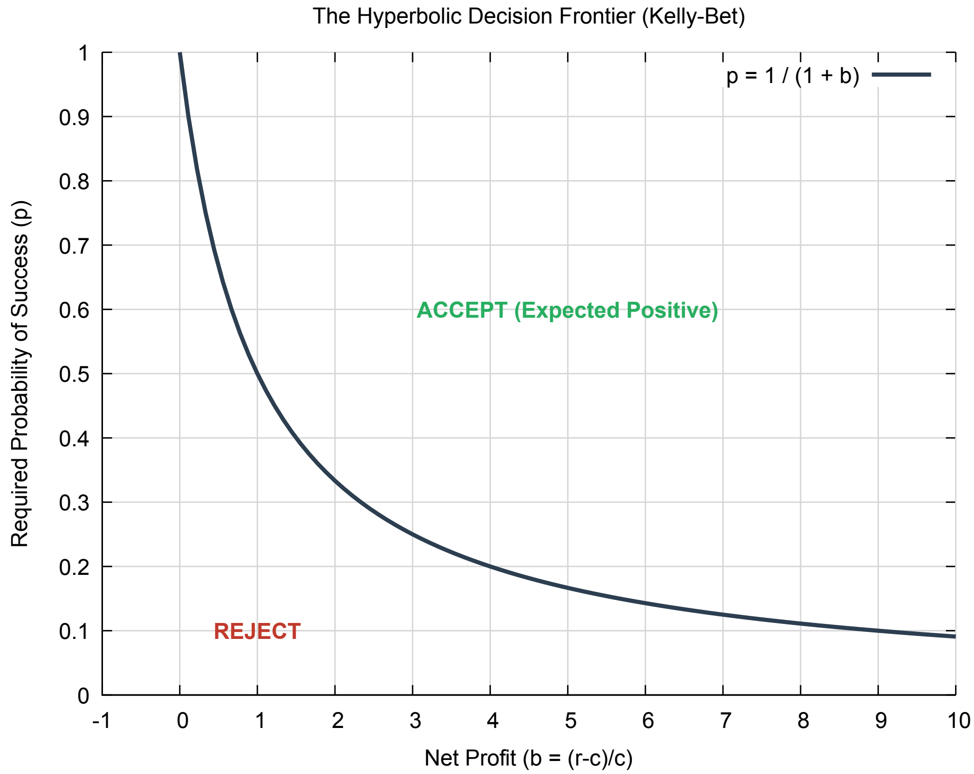

Figure 1: The Hyperbolic Decision Frontier. The curve $p = 1/(1+b)$ represents the zero-growth boundary. Areas in green represent the "Accept" region where the estimated success probability p satisfies the Kelly-viability constraint.

4 Deriving the Pareto Frontier

Substituting the Bayesian expected probability into the Kelly constraint - after minor algebraic simplifications - reveals the simple, elegant relation that governs the Pareto Frontier for accepting or rejecting a deal, represented as a critical ratio: $b > \beta/\alpha$

5 Core Insight for the Communication

The profitable area in figure 1 is bounded by a hyperbola with asymptotes at $b = -1$ and $p = 0$. Plainly stated: for a positive outcome, all it takes for a rational player is that the net profit (b) stay above the ratio of frictions to successes or "cons to pros” (β/α).

Acknowledgments

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References

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